37. How fast must a 54g tennis ball travel in order to have a de Broglie wavelength that is equal to that of a photon of green light 5400Å?

38. For each of the following, give the sub level designation, the allowable m values and the number of orbitals

i) n = 4, l =2, ii) n =5, l = 3 iii) n=7, l=0

39. Give the electronic configuration of Mn^{2+} and Cr^{3+}

40. Describe the Aufbau principle

41. An atom of an element contains 35 electrons and 45 neutrons. Deduce

i) the number of protons

ii) the electronic configuration for the element

iii) All the four quantum numbers for the last electron

42. Show that the circumference of the Bohr orbit for the hydrogen atom is an integral multiple of the de Broglie wave length associated with the electron revolving arround the nucleus.

43. Calculate the energy required for the process.

$$He^{+}_{(g)} \rightarrow He^{2+}_{(g)} + e^{-}$$

The ionisation energy for the H atom in its ground state is - 13.6 ev $atom^{-1}$.

44. An ion with mass number 37 possesses unit negative charge. If the ion contains 11.1% more neutrons than electrons. Find the symbol of the ion.

45. The Li^{2+} ion is a hydrogen like ion that can be described by the Bohr model. Calculate the Bohr radius of the third orbit and calculate the energy of an electron in 4^{th} orbit.

46. Protons can be accelerated in particle accelerators. Calculate the wavelength (in Å) of such accelerated proton moving at 2.85×10^{8} ms^{-1} (the mass of proton is 1.673×10^{-27} Kg).

47. What is the de Broglie wavelength (in cm) of a 160g cricket ball travelling at 140 Km hr^{-1}.

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